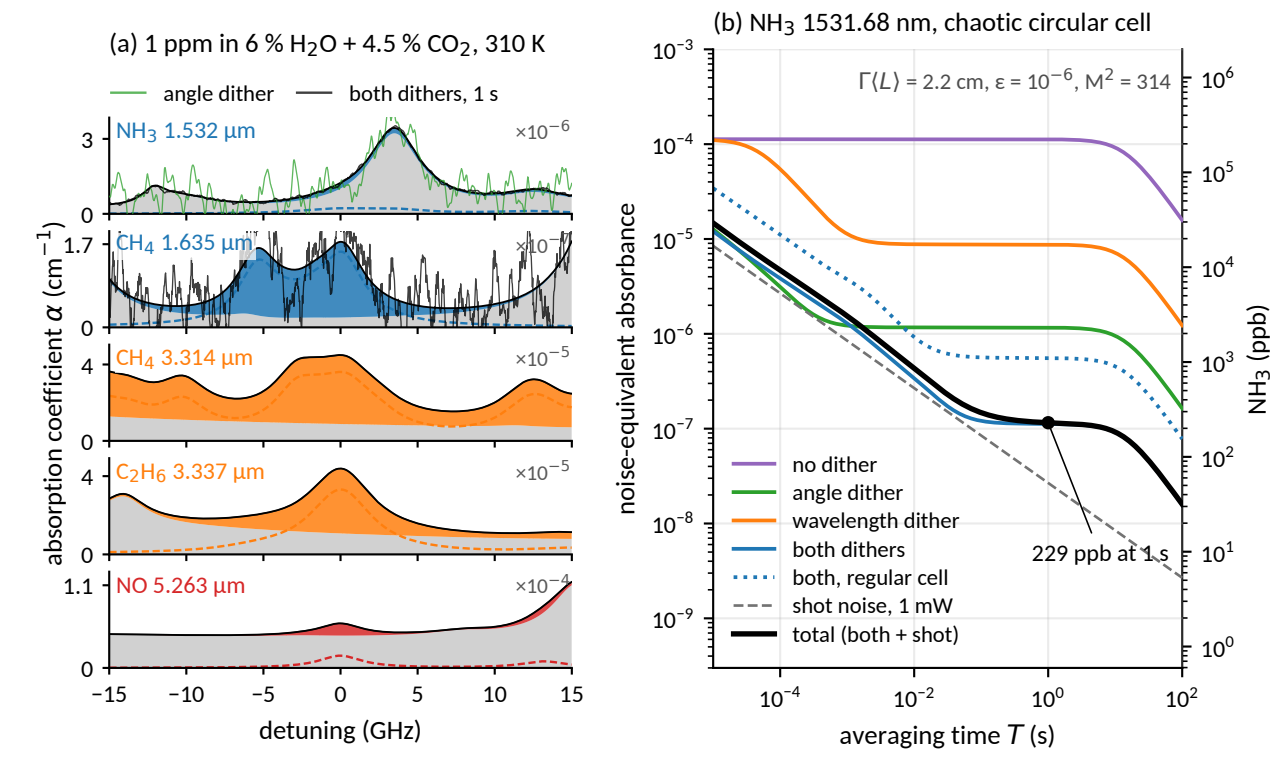

(a) 1 ppm in 6 % H2O + 4.5 % CO2, 310 K
angle dither
both dithers, 1 s
NH3 1.532 µm
CH4 1.635 µm
CH4 3.314 µm
C2H6 3.337 µm
NO 5.263 µm
absorption coefficient α (cm−1)
detuning (GHz)
(b) NH3 1531.68 nm, chaotic circular cell
Γ⟨L⟩ = 2.2 cm, ε = 10−6, M2 = 314
noise-equivalent absorbance
NH3 (ppb)
averaging time T (s)
no dither
angle dither
wavelength dither
both dithers
both, regular cell
shot noise, 1 mW
total (both + shot)
229 ppb at 1 s